Shifting Attention to Accuracy Can Reduce Misinformation Online

Replication of Pennycook et al. (2021, Nature)

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Why Do People Share Misinformation?

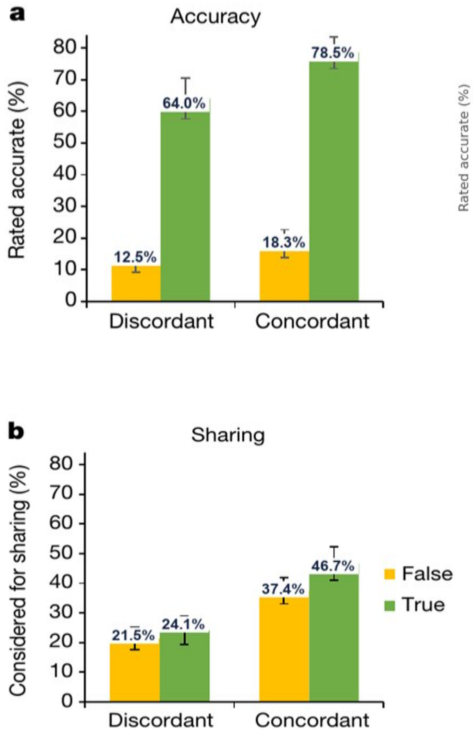
Three possible reasons based on the paper:

- Confusion
- Preference
- Inattention

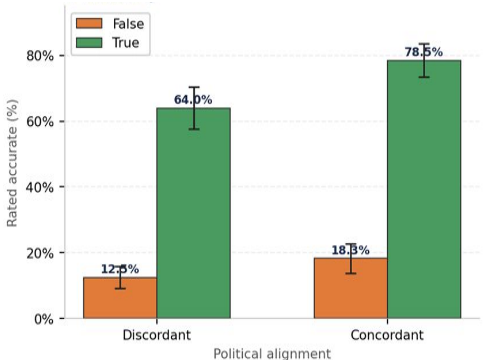
Study 1: There is no difference in willingness to share between true and false headlines

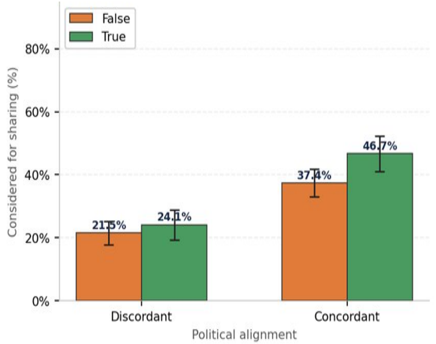
1,005 US adults on MTurk rated 36 politically-balanced headlines (18 true, 18 false) for either accuracy or willingness to share (a between-subjects experiment)

Paper Results



Python Replication





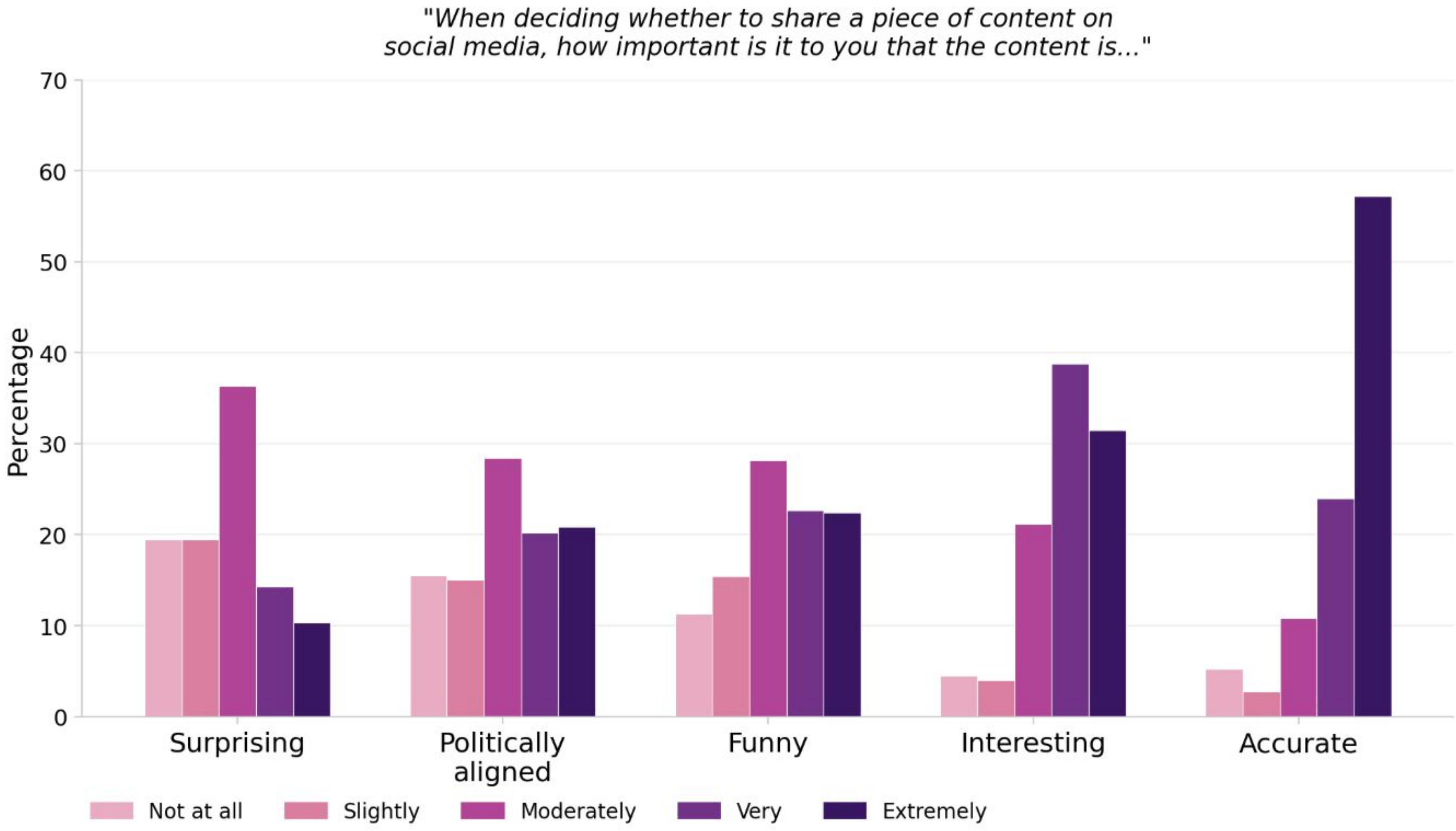
Gaps identified (Paper vs Stata vs Python)

#	Gap	Risk
1	SE clustering changed from participant-only (pre) to two-way post pre-registration	Justified and explained in paper but just undisclosed in code.
2	2 of 3 exclusion filters never explicitly coded	Stata log confirms **didnt_finish** encodes all three groups
3	Sample size: 1,015 stated; both Stata and Python yield 1,005	Paper's n=1,015 reproducible from neither code nor execution log

Design Limitation

DV (Dependent variable) measures declared sharing intent, not observed platform behavior. Effect magnitude may not transfer to real-world sharing. This was acknowledged by authors, not quantified.

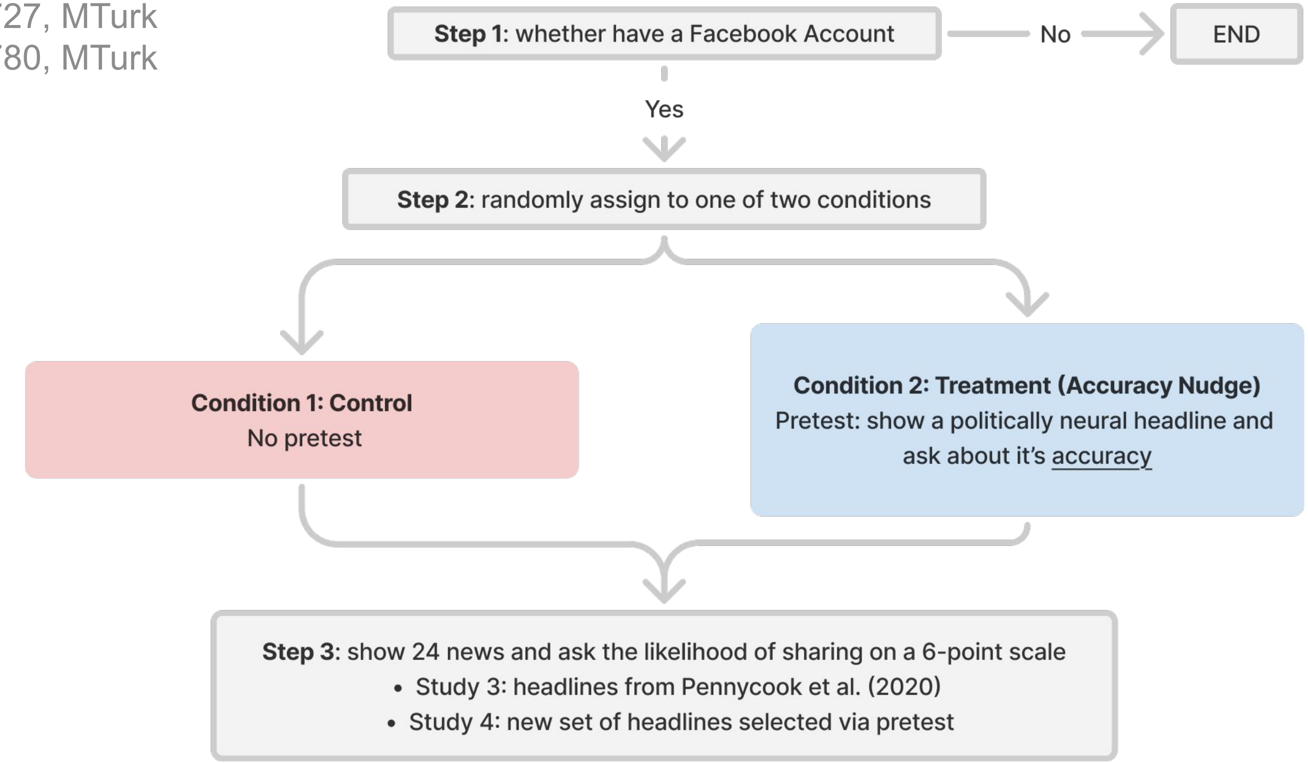
Study 2: People do care about accuracy



Study 3-4

Does subtly shifting attention to accuracy increase the veracity of the news people are willing to share?

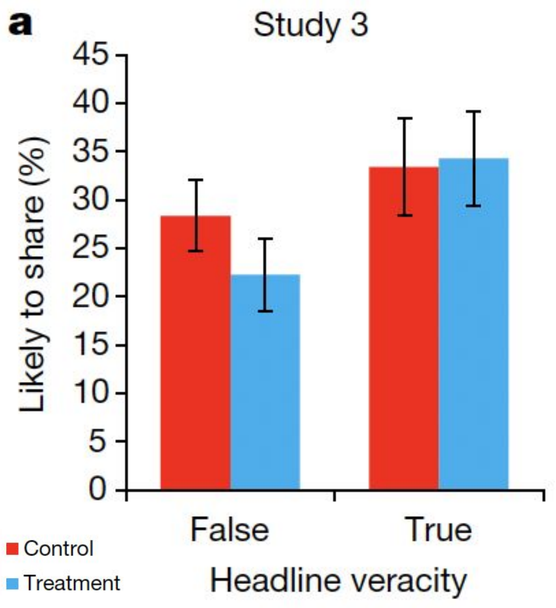
Study 3: n = 727, MTurk
Study 4: n = 780, MTurk



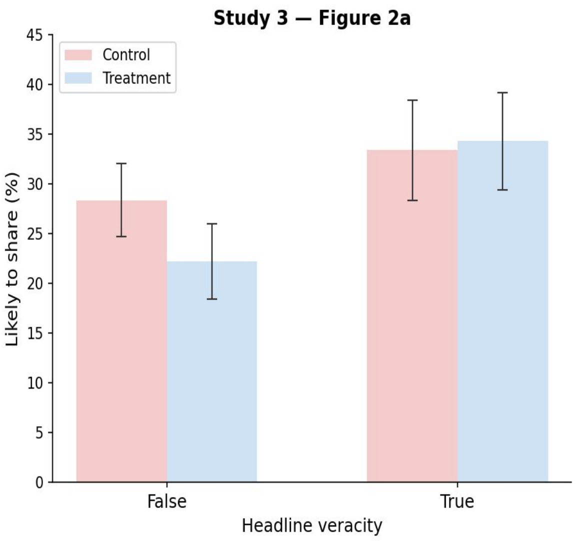
Shifting attention to accuracy can reduce misinformation online (Pennycook et al., 2021)

Study 3-4 | Accuracy nudges improve sharing discernment

Original Figures



Replication Figures



Observations

Study 3

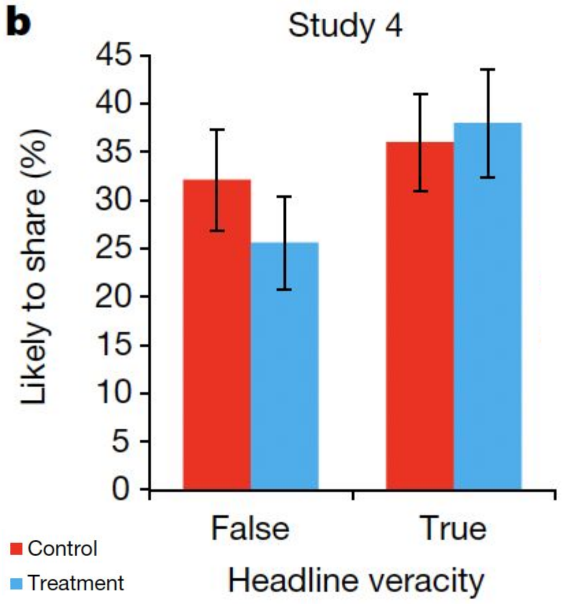
In the treatment condition, willingness to share false headlines decreased while willingness to share true headlines remained unchanged

The replicated figure shows ***identical*** *results to the original figure*

Study 3-4

Accuracy nudges improve sharing discernment

Original Figures



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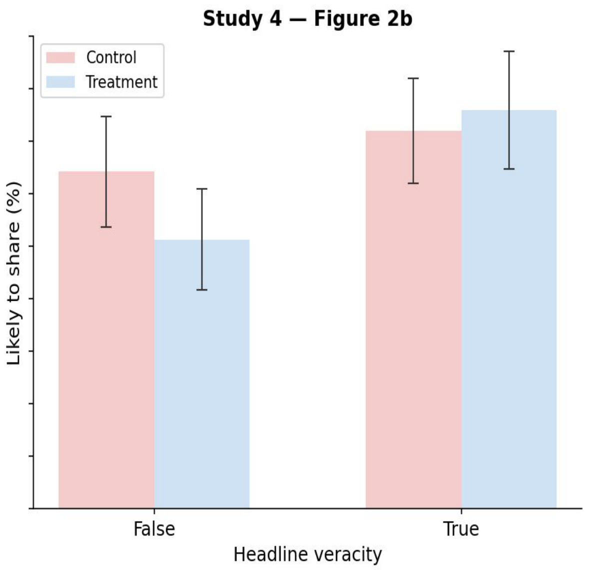

Observations

Study 4

In the treatment condition, willingness to share false headlines decreased while willingness to share true headlines remained unchanged

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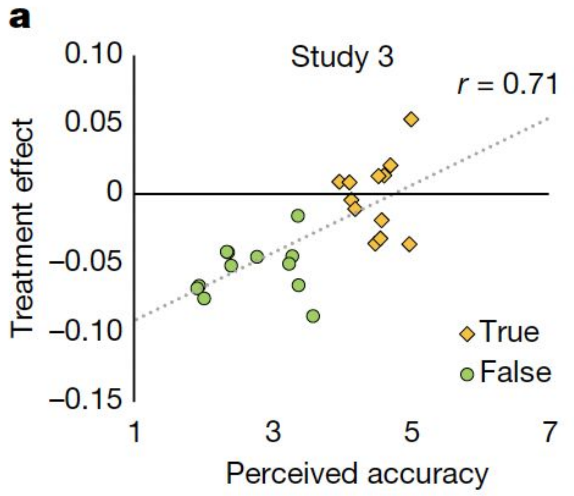
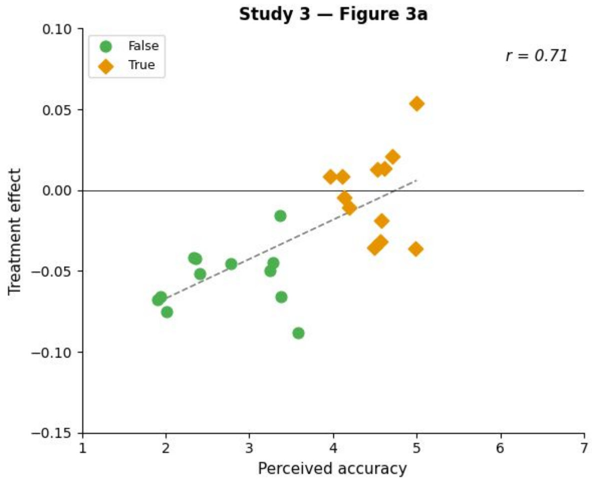
Study 3-4 | Accuracy nudges improve sharing discernment

Original Figures



Replication Figures



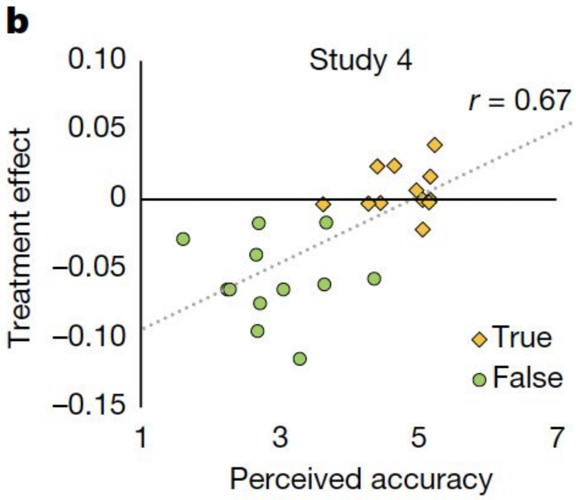
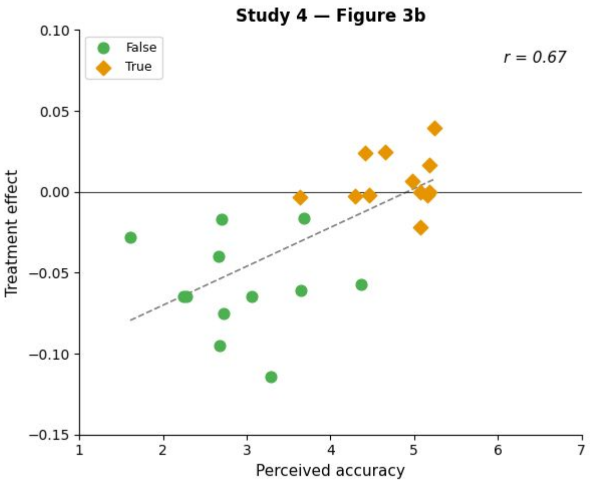
Observations

Study 3

The reduction in sharing is largest for headlines perceived as least accurate, while sharing of more accurate headlines remains largely unchanged

*The replicated figure shows **identical** results to the original figure, with the correlation values also matching exactly*

Study 3-4 | Accuracy nudges improve sharing discernment

Original Figures	Replication Figures	Observations
		**Study 4**

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Study 3-4

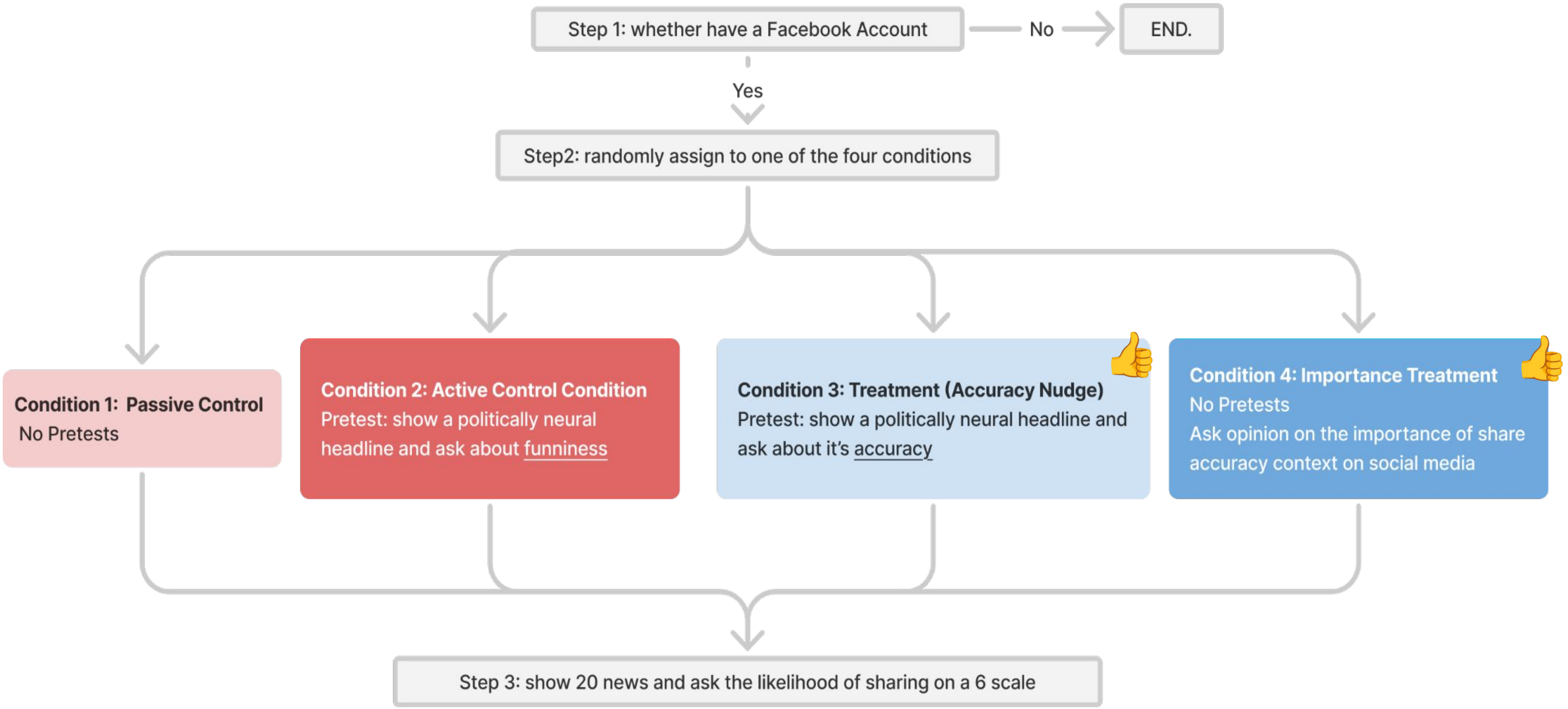
Does subtly shifting attention to accuracy increase the veracity of the news people are willing to share?

Discrepancies Between PAP, Original Paper, and Authors' Stata Code				
Analysis	**PAP**	**Paper**	**Stata**	**Python**
Bayesian t-test (AccImp & AccImp_Friends): BF_{10} = 0.063 and BF_{10} = 0.095; no significant difference in accuracy importance between conditions **(Paper p.592)**	O	O	X	O
Chi-square test (sharer balance): $\chi^2(1) = 0.156$, $p = .69$ (Study 3); $\chi^2(1) = 0.988$, $p = .32$ (Study 4); socialmedia_chk does not differ by condition **(Paper p.597)**	O	O	X	O
Simple effects by partisanship × concordance: Condition effect on false headlines for each combination of participant partisanship and headline concordance **(Supplementary Table S3 caption)**	O	O	X	O
CRT (Cognitive Reflection Test) as moderator of treatment effect: 3-way interaction condition × real × CRT; does treatment effect vary by CRT?	**O**	**X**	**X**	O

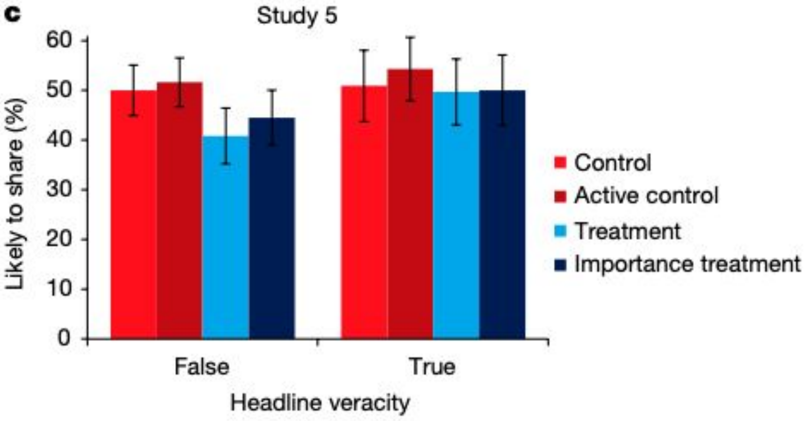
Shifting attention to accuracy can reduce misinformation online (Pennycook et al., 2021)

Study 5: Nudge Effect on Misinformation Sharing

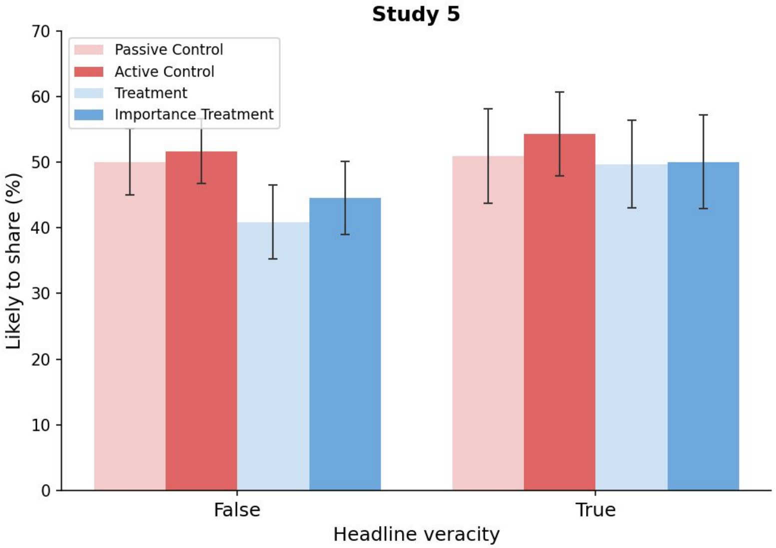
Population: n = 1,268 American individuals from Lucid
Chi-square test conducted to verify that the proportion of sharers was balanced across all four conditions



Paper Result:



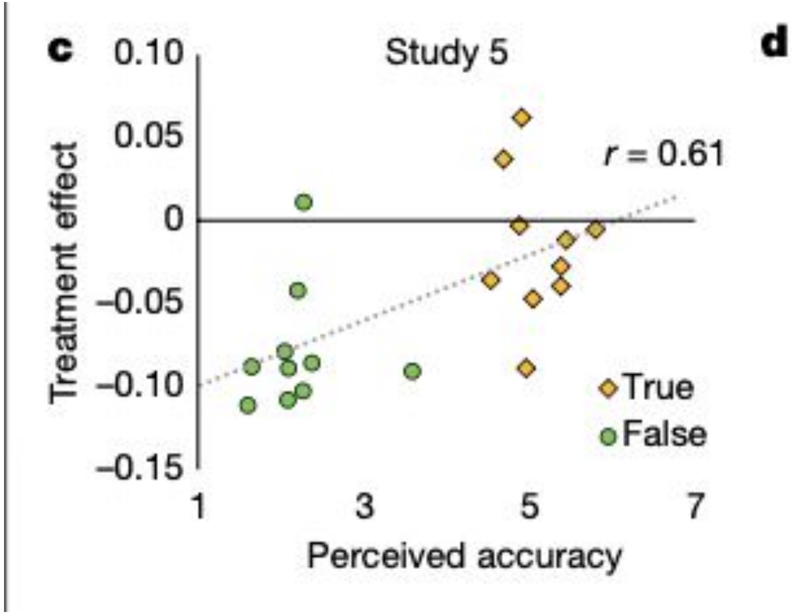
Replication Result:



Condition	False headlines	True headlines	True − False gap
Passive Control	50.0%	50.9%	+0.9%
Active Control	51.6%	54.3%	+2.7%
Treatment	**40.9%**	**49.7%**	**+8.8%**
Importance Treatment	**44.6%**	**50.0%**	**+5.4%**

- both a <u>subtle</u> accuracy prime and a more <u>direct</u> accuracy-importance prompt both increase sharing discernment.
- reducing willingness to share false headlines, while leaving willingness to share true headlines mainly unchanged.
- **Interesting Question:** Does rating funniness make people more likely to share any content?

Paper Result:



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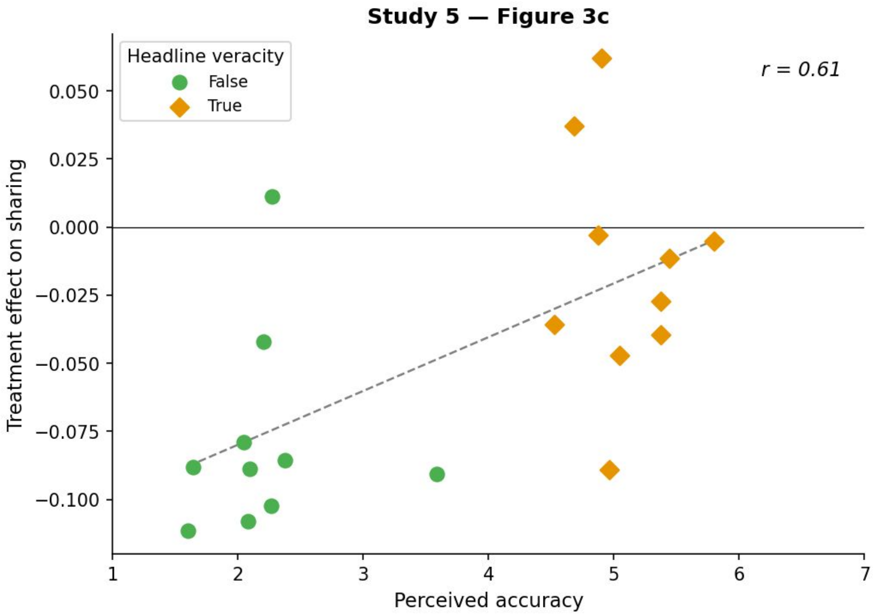
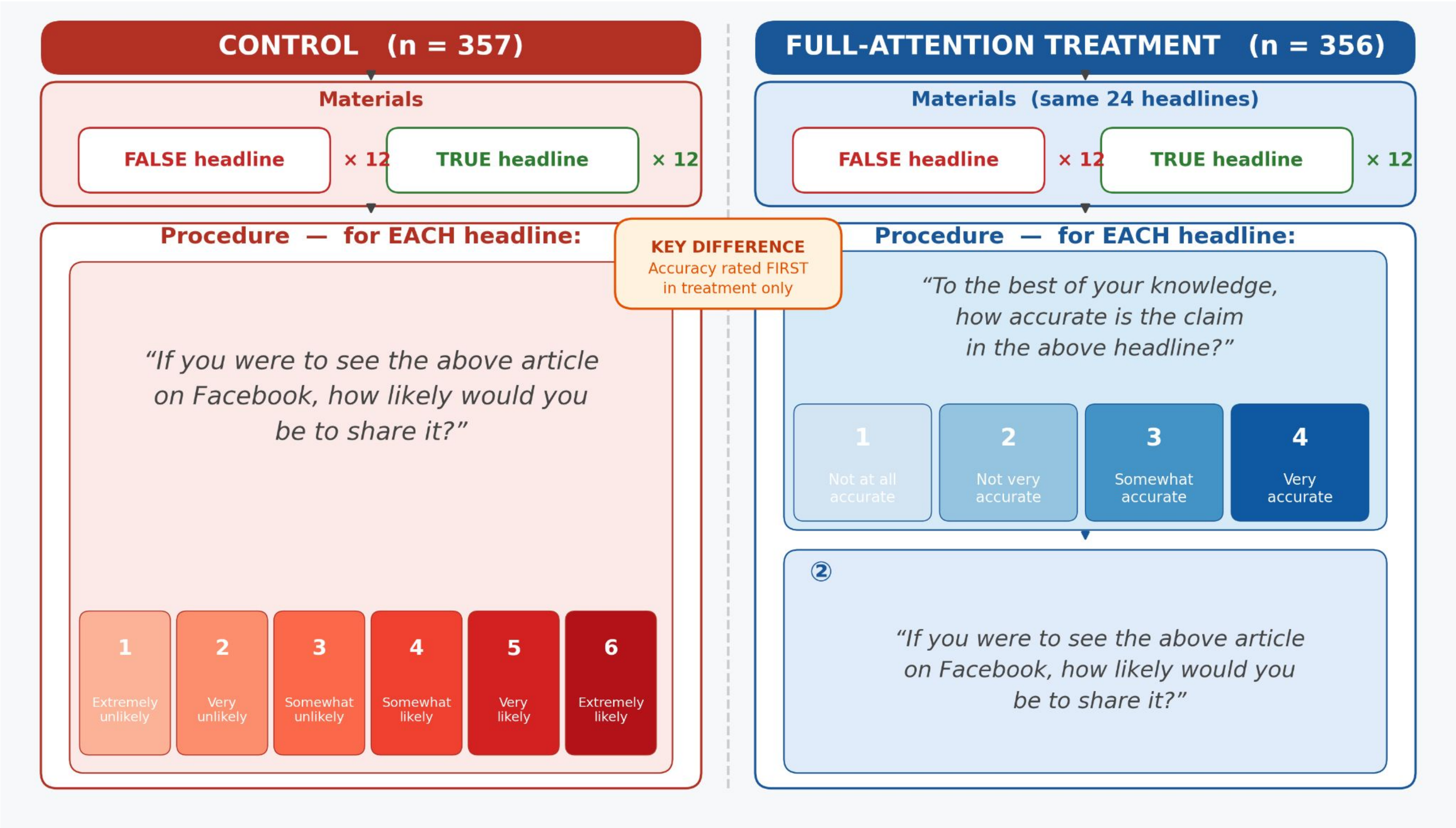
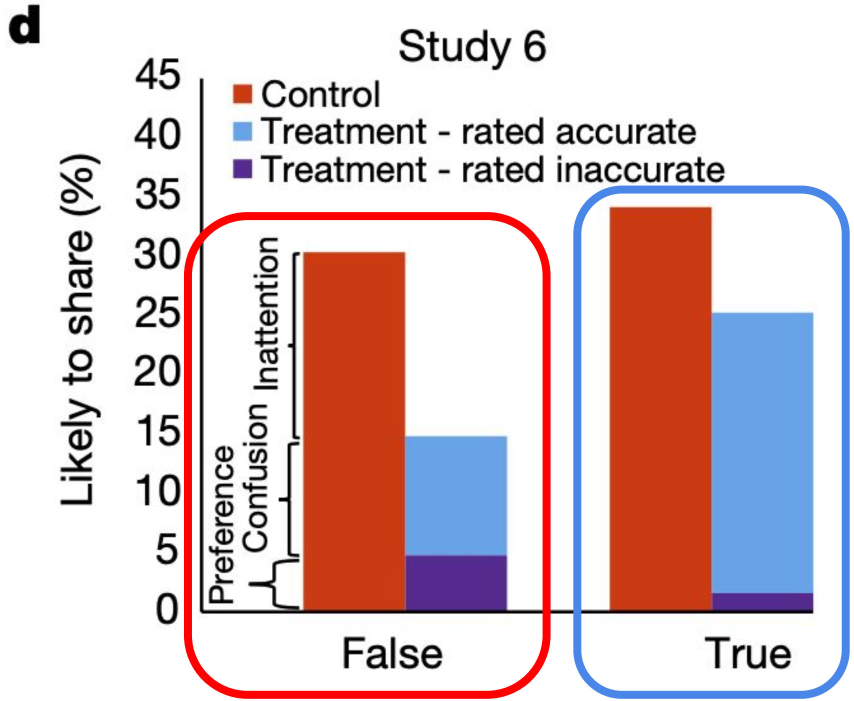


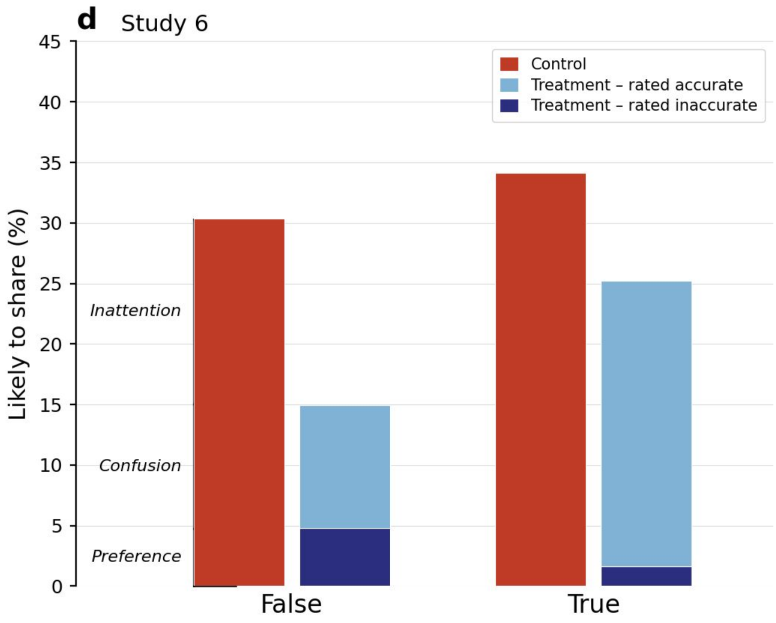
Study 6 | Shifting attention to accuracy is the ***mechanism*** behind this effect



Study 6 | Inattention accounts for >50% of false headlines sharing
Reduced Propensity to Share True Headlines Is Left Undiscussed



Paper Result



Our Replication

Storyline of the Paper

How shifting attention to accuracy reduces misinformation sharing

1 Disconnect between sharing and accuracy

Studies 1–2

- **Study 1:** People distinguish true from false when judging accuracy, but veracity matters much less for sharing.
- **Study 2:** People still report that accuracy is the most important factor when deciding what to share.

Key insight: People value accuracy, but sharing decisions are partly disconnected from it.

2 Priming accuracy improves sharing

Studies 3–5

- A subtle accuracy prompt shown before the sharing task makes people more discerning.
- The effect mainly reduces sharing of false headlines, with little change for true headlines.

Key insight: Making accuracy salient improves sharing quality.

3 Attending to accuracy as the mechanism

Study 6

- Participants judge each headline's accuracy immediately before deciding whether to share it.
- This separates confusion, preference, and inattention, showing that inattention explains the largest share.

Key insight: The main problem is failing to attend to accuracy.

Overall conclusion

A simple accuracy nudge can reduce misinformation sharing because many people care about truth, but do not naturally focus on accuracy when deciding what to share.